SPECIAL: Zero-Shot Hyperspectral Image Classification With CLIP

Li Pang, Jing Yao, *Senior Member, IEEE*, Kaiyu Li, Jun Zhou, *Fellow, IEEE*, Deyu Meng, and Xiangyong Cao

***Abstract*—Hyperspectral image (HSI) classification aims to categorize each pixel in an HSI into a specific land cover class, which is crucial for applications such as remote sensing, environmental monitoring, and agriculture. Although deep learning (DL)-based HSI classification methods have achieved significant advancements, existing methods still rely on manually labeled data for training, which is both time-consuming and labor-intensive. To address this limitation, we introduce a novel zero-shot hyperspectral image classification framework based on CLIP (SPECIAL), aiming to eliminate the need for manual annotations. The SPECIAL framework consists of two main stages: 1) contrastive language–image pretraining (CLIP)-based pseudo-label generation (PLG) and 2) noisy label learning (NLL). In the first stage, HSI is spectrally interpolated to produce red, green, and blue (RGB) bands. These bands are subsequently classified using CLIP, resulting in noisy pseudo-labels that are accompanied by confidence scores. To improve the quality of these labels, we propose a scaling strategy that fuses predictions from multiple spatial scales. In the second stage, spectral information and a label refinement technique are incorporated to mitigate label noise and further enhance classification accuracy. Experimental results on three benchmark datasets demonstrate that SPECIAL achieves competitive and often superior performance in zero-shot HSI classification, showing its potential for more practical applications. The code is available at https://github.com/LiPang/SPECIAL**

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Li Pang is with the School of Mathematics and Statistics and the Ministry of Education Key Laboratory of Intelligent Networks and Network Security, Xi'an Jiaotong University, Xi'an, Shaanxi 710049, China (e-mail: 2195112306@stu.xjtu.edu.cn).

Jing Yao is with the State Key Laboratory of Remote Sensing and Digital Earth, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: jasonyao92@gmail.com).

Kaiyu Li and Xiangyong Cao are with the School of Computer Science and Technology and the Ministry of Education Key Laboratory for Intelligent Networks and Network Security, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: likyoo.ai@gmail.com; caoxiangyong@xjtu.edu.cn).

Jun Zhou is with the School of Information and Communication Technology, Griffith University, Nathan, QLD 4111, Australia (e-mail: jun.zhou@griffith.edu.au).

Deyu Meng is with the School of Mathematics and Statistics and the Ministry of Education Key Laboratory of Intelligent Networks and Network Security, Xi'an Jiaotong University, Xi'an, Shaanxi 710049, China, and also with the Pazhou Laboratory (Huangpu), Guangzhou, Guangdong 510555, China (e-mail: dymeng@mail.xjtu.edu.cn).

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I. Introduction

HYPERSPECTRAL imaging captures detailed spectral information at multiple wavelengths for each pixel, extending far beyond the capabilities of traditional optical imaging modes. The rich spectral information in the acquired hyperspectral images (HSIs) enables the identification of materials through their unique spectral signatures, leading to various applications such as mineral exploration [1], [2], [3], environmental monitoring [4], [5], and land cover and land use classification [6], [7].

Among these applications, HSI classification, which involves assigning each pixel to a specific land cover class, has become a popular research area in recent years. Earlier methods often involve manual feature extraction and traditional machine learning (ML) to identify relevant features from spectral data [8], [9]. However, these methods suffer from biased prior knowledge and lack the ability to capture nonlinear relationships under complex scenes. With the significant progress in deep learning (DL) techniques, numerous studies have explored its application to HSI classification. Convolutional neural networks (CNNs) [10], [11], [12] and Transformers [13], [14], [15] have been widely employed to enhance the classification performance. More recently, the Mamba [16] architecture, which benefits from both high computational efficiency and long-range modeling capabilities, has also gained popularity in hyperspectral classification [17], [18], [19].

Although these models have demonstrated impressive performance, their effectiveness still heavily depends on complete and accurate manually labeled data. This dependency becomes a major obstacle for practical HSI analysis for three reasons. First, pixel-wise annotation requires domain expertise because the class identity of a hyperspectral pixel is often determined by both its visual appearance and its spectral signature. Second, real-world remote sensing often follows an open-scene setting: when a model is deployed in a new geographical region, the land-cover categories may differ from those in the source data, so identical categories do not need to be annotated again, but novel region-specific categories generally do. Third, the spectral response of the same land-cover type can change across sensors, illumination conditions, seasons, and atmospheric environments, leading to spectral

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shifts that degrade the transferability of supervised models. Therefore, both category-level openness and sensor-induced spectral variation can repeatedly require new annotations, which motivates the study of zero-shot HSI classification.

Recently, CLIP [20], a deep model trained to align images and text in a shared embedding space, has gained increasing popularity in open-vocabulary semantic segmentation [21], [22], [23]. In these approaches, compact image and text features are generally first obtained through the visual and text encoders, respectively. Then, the image features are upsampled to the original spatial size, allowing each pixel to be classified by measuring the similarity between its features and a set of text features. Several studies have also attempted to apply the CLIP model to semantic segmentation in remote sensing [24], [25], demonstrating promising performance in open-vocabulary settings. Very recently, DiffCLIP [26] employs CLIP for few-shot hyperspectral classification. However, the application of the CLIP model for zero-shot HSI classification remains unexplored.

To further explore the potential of visual and language models for intelligent hyperspectral interpretation in real-world scenarios, in this article, we propose a novel zero-shot hyperspectral image classification framework based on CLIP (SPECIAL). For hyperspectral data in new scenes, we first spectrally interpolate the original HSI to simulate red, green, and blue (RGB) data that is compatible with on-the-shelf CLIP, and then utilize the latter to perform unsupervised classification, obtaining pseudo-labels and corresponding confidence scores for each pixel. We also introduce the multiscale mechanism in this process to improve the quality of the pseudo-labels generated by CLIP. Next, we consider incorporating spectral information to boost the classification performance further. Specifically, we leverage the pseudo-labels provided by CLIP to guide the training of an HSI classification network. Moreover, considering that there is substantial noise in the pseudo-labels, we propose a label refinement strategy that utilizes Gaussian mixture models (GMMs). By identifying high-confidence and low-confidence samples and modeling their distributions, we are able to generate soft labels of additional training samples that can effectively refine the training process. Experiments on three datasets demonstrate that our approach achieves competitive and often superior performance compared with existing unsupervised classification approaches, one-shot HSI classification baselines, and recent zero-shot HSI classification methods. To summarize, the main contributions of this article are as follows.

1) We propose a novel zero-shot HSI classification framework, termed **SPECIAL**, which enables HSI classification without manually annotated labels by transferring CLIP's open-vocabulary semantic prior into the hyperspectral domain.
2) To improve the recognition accuracy of CLIP models for objects of varying sizes, a resolution scaling (RS) strategy, which fuses predictions under different image resolutions, is proposed to improve the quality of the pseudo-labels provided by CLIP.
3) To further improve the performance of HSI classification, we propose a noise-robust framework that incorporates both hyperspectral information and CLIP prior, in which three training subsets are dynamically sampled to alleviate the issue of label noise.

The remainder of this article is structured as follows. Section II reviews related work on HSI classification and CLIP-based segmentation. Section III details our proposed SPECIAL framework, including pseudo-label generation (PLG) and noisy label learning (NLL). Section IV presents extensive experimental results and ablation studies on three benchmark datasets. Finally, Section V concludes the article and discusses future work directions.

II. Related Work

A. HSI Classification

HSI classification methods are diverse and evolving, spanning from traditional ML approaches to modern DL methods. Early ML-based methods relied on manually crafted features from the spectral signatures and then applied classical classifiers. For example, support vector machines (SVMs) [8], [9], [27] and random forests (RFs) [28], [29], [30] were a popular choice for HSI classification due to their effectiveness in high-dimensional space. These traditional methods often incorporated dimensionality reduction [e.g., principal component analysis (PCA)] [31], [32] to handle the high spectral dimensionality. While such approaches are straightforward and require little data, they generally struggle to capture complex nonlinear relationships and spatial context in the imagery.

In the past decade, DL methods have dramatically advanced HSI classification performance. CNNs were among the first DL models applied to HSI, initially processing the data as either 1-D spectral sequences or 2-D spatial images [11], [12]. Beyond CNNs, Transformer-based models have entered HSI analysis recently. They treat the HSI as a sequence of tokens (patches or pixels) and use self-attention to model long-range dependencies. Yao et al. [14] introduced a vision transformer for land-cover classification that integrates spectral and spatial tokens, and Zhao et al. [15] applied a transformer with separable convolutions to efficiently handle HSI data. These transformer models can capture global context better than local CNN filters, albeit with higher data requirements. Another emerging approach is the Mamba architecture, based on state-space models, which achieves linear time complexity with the ability to model long-range interactions. Mamba-based HSI classifiers [17], [18], [19] have shown promising results. They effectively encode the HSI as a sequence and use learned state transitions to capture global context, offering a memory-efficient alternative to Transformers. Overall, DL methods significantly outperform traditional methods by automatically learning expressive spectral–spatial features. However, they usually demand a considerable amount of labeled data for training and can struggle when labels are scarce or when facing new sensor types or acquisition conditions.

To mitigate the reliance on labeled data, researchers have explored pretraining and domain-adaptive techniques for HSI classification. Self-supervised representation learning and pretraining on auxiliary tasks have been used to initialize models that can then be fine-tuned with few labels. For example,

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Lee et al. [33] pretrained models on a large corpus of unlabeled HSI data (and even cross-domain data) to learn general features before fine-tuning on a target HSI classification task. Contrastive learning methods have also been employed: Guan and Lam [34] propose a cross-domain contrastive learning framework that learns sensor-invariant features by pulling together representations of the same pixel under different augmentations and pushing apart others. Li et al. [35] combine supervised contrastive learning with unsupervised domain adaptation to transfer a model from a source domain to a target domain with minimal labels. These approaches improve generalization to new scenes or sensors but typically still assume at least some labeled data in the target domain for fine-tuning or validation. Despite these advances, all the above methods still require some human annotations. In scenarios where obtaining labels is extremely difficult (e.g., new hyperspectral sensors or exotic environments), it becomes attractive to explore zero-shot learning approaches, which we discuss later. This has motivated our work in eliminating the need for any task-specific labeled data by leveraging prior knowledge from a pretrained vision–language model (VLM) [36], [37].

B. CLIP-Based Open Vocabulary Semantic Segmentation

CLIP [20] learns to align images with their corresponding text descriptions in the feature space. CLIP employs a dual-encoder architecture, in which an image encoder and a text encoder are jointly trained with a contrastive objective so that matched image–text pairs are pulled close, and mismatched pairs are pushed apart in a shared feature space. Once trained on large-scale web data, CLIP can directly perform zero-shot image classification by comparing the visual feature of an input image with text embeddings of category names expressed in natural language prompts, thereby enabling recognition over an essentially open vocabulary without task-specific fine-tuning. Motivated by this impressive zero-shot capability, a growing body of work attempts to transfer CLIP's knowledge of visual concepts from image-level classification to dense prediction tasks, giving rise to CLIP-based open-vocabulary semantic segmentation, where each pixel (or patch) is assigned a label drawn from an unrestricted set of categories.

MaskCLIP [38] makes an early attempt to adapt CLIP to pixel-level segmentation by reusing the pretrained visual backbone and converting patch-level representations into dense predictions guided by text features, thus extracting pseudo labels for each spatial location without the need for additional human annotations. Building upon this idea, subsequent methods such as SCLIP [39], GEM [40], and ClearCLIP [41] further explore how to more effectively exploit the internal structure of CLIP. In particular, they typically refine the attention mechanisms, redesign the interaction between visual and textual tokens, and introduce lightweight architectural modifications (e.g., decoder heads or feature fusion modules) to enhance the ability of CLIP features to capture fine-grained spatial details, leading to improved performance on dense prediction benchmarks under the open-vocabulary setting.

Recently, SegEarth-OV [25] extends CLIP-based open-vocabulary segmentation to the field of remote sensing. It incorporates a novel feature upsampler, namely, SimFeatUp, together with a global bias alleviation strategy to mitigate the sensitivity of CLIP to low-resolution, small-scale objects that are common in aerial and satellite imagery. This design effectively strengthens the representation of subtle structures such as narrow roads or small buildings, and demonstrates significant improvements in practical applications including building extraction and road detection. In addition, DiffCLIP [26] has recently explored the use of CLIP for few-shot HSI classification, showing that CLIP's language-aligned visual features can be beneficial even when only a handful of labeled HSI samples are available. However, despite these advances, zero-shot HSI classification based on CLIP remains largely unexplored, and there is still a lack of dedicated frameworks that fully exploit CLIP's open-vocabulary capabilities in the hyperspectral domain.

III. Method

The overall framework of SPECIAL is shown in Fig. 1. The pipeline is composed of two sequential, logically connected stages: CLIP-based PLG and NLL. In the PLG stage, we extract RGB bands from the HSI through interpolation. Subsequently, a CLIP-based open-vocabulary segmentation model, SegEarth-OV [25], classifies the RGB pixels to produce initial pseudo-labels and confidence scores. To mitigate spatial inaccuracies caused by varying object sizes, we introduce an RS strategy that fuses multiscale predictions. However, despite the improved spatial alignment provided by the RS strategy, the inherent domain gap between the interpolated RGB proxy images and the rich hyperspectral data inevitably introduces substantial label noise.

To address this persistent label corruption, the NLL stage employs a robust learning paradigm divided into two distinct, sequentially dependent phases: a warmup phase and a label refinement phase. The warmup phase is essential to safely initialize the spectral classifier and distill the coarse semantic prior from the RGB proxy into the pure hyperspectral domain. Because remote sensing scenes suffer from extreme class imbalance and severe label noise, directly training on all pseudo-labels would cause the model to rapidly overfit. Therefore, during the warmup phase, we train a MambaHSI [18] network—chosen for its efficient long-range modeling capabilities—using a class-balanced sampling strategy weighted by the initial CLIP confidence scores. This controlled initialization is the fundamental prerequisite for the next step: it ensures the network can output meaningful best versus second best (BvSB) [42], [43] confidence distributions. Without this warmup phase, the predicted scores would be entirely random, causing the subsequent noise modeling to fail entirely. Following this, the label refinement phase leverages these initialized, meaningful scores to dynamically correct the noisy labels. Specifically, we categorize the predictions into confident and hard sets based on the BvSB distribution and model these sets using GMMs. We then calculate the probability density of each sample under the class-specific GMM distributions. These densities are normalized to yield soft pseudo-labels, which replace the noisy labels to further optimize the classification performance.

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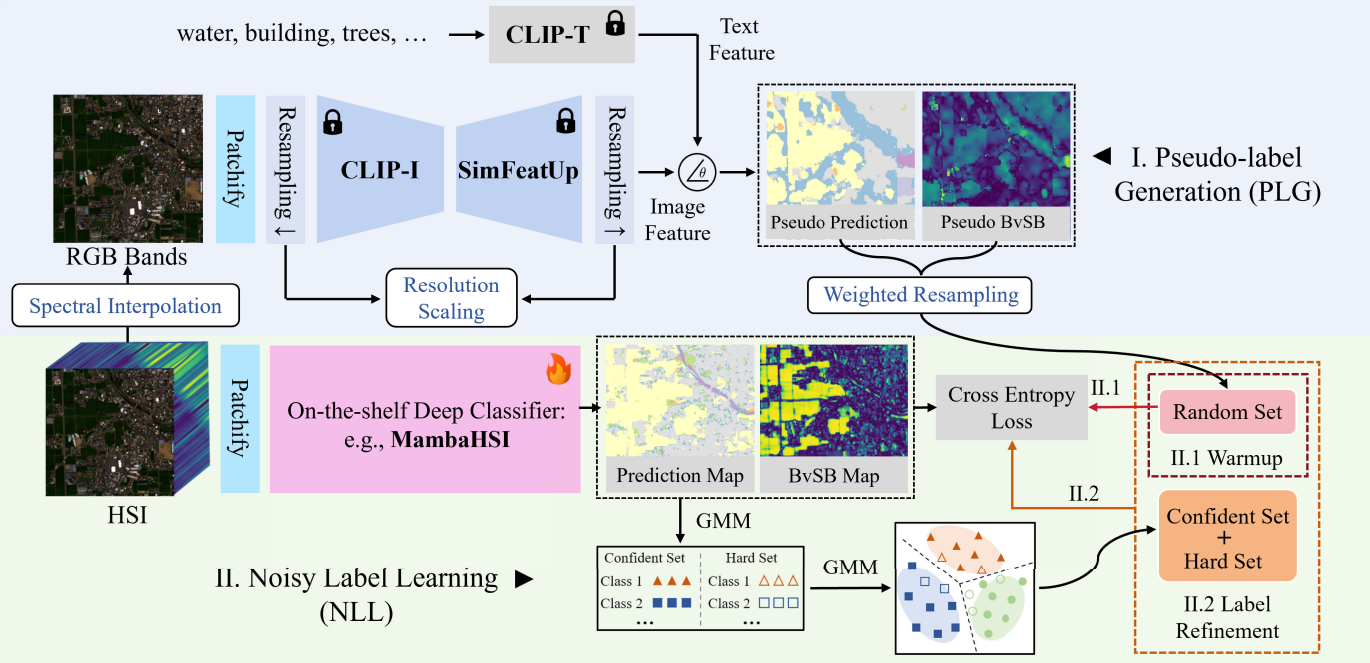


Fig. 1. Overall framework of our proposed SPECIAL. The proposed framework consists of two stages: CLIP-based PLG and NLL. In the PLG stage, CLIP classifies interpolated RGB bands, generating pseudo-labels with confidence scores, while NLL further improves classification accuracy by incorporating spectral information with a label refinement strategy.

From a high-level perspective, SPECIAL fundamentally transfers the open-vocabulary semantic prior encoded in CLIP into the hyperspectral domain. The PLG stage harvests category-aware spatial supervision, while the NLL stage systematically models and filters the domain-shift noise via GMM-based refinement. This decoupled design ensures our framework is highly modular: more powerful CLIP variants or HSI backbones can be plugged in without redesigning the overall training paradigm. A detailed description of each component is provided below.

It should be emphasized that SegEarth-OV is used as the open-vocabulary pseudo-label generator in the PLG stage, while the contribution of SPECIAL lies in adapting this generator to the HSI classification problem and coupling it with a spectral noise-robust learning stage. Specifically, the PLG stage aligns HSI bands with physical RGB wavelengths, estimates pixel-wise pseudo-label reliability through BvSB confidence, and improves dense predictions through the proposed RS strategy. The subsequent NLL stage then transfers these noisy RGB-domain semantic cues to the original high-dimensional hyperspectral domain, where spectral distributions are explicitly modeled for label refinement.

A. CLIP-Based PLG

In this stage, we first transform the original HSI into a three-channel RGB-like image that can be directly processed by VLMs. Specifically, we linearly interpolate the spectral bands around 655, 553, and 451 nm, which approximately correspond to the RGB channels, respectively. This yields a false-color RGB image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$ that preserves the main spatial structures and semantic cues of the HSI while remaining compatible with CLIP-based architectures. This conversion establishes a connection between the hyperspectral domain and the natural-image domain where CLIP [20] is trained.

Next, we adopt the training-free open-vocabulary segmentation framework SegEarth-OV [25] to obtain pixel-wise semantic scores and confidence maps from $\mathbf{I}$. SegEarth-OV employs a ViT-based CLIP image encoder (CLIP-I) and a CLIP text encoder (CLIP-T) to map images and category names into a shared embedding space.

We choose SegEarth-OV as the default pseudo-label generator because it is specifically designed for remote sensing scenes and provides stronger dense predictions than generic CLIP-based segmentation models under large-scale aerial or satellite imagery. Its intermediate-feature projection, SimFeatUp upsampler, and global-bias alleviation are particularly useful for remote sensing images, where small objects, thin structures, and large homogeneous regions coexist. Since the proposed NLL stage is trained from these pseudo-labels, higher pseudo-label quality is directly beneficial to our framework. This tendency is also supported by the ablation study in Section IV-D, where improved pseudo-labels from SegEarth-OV and the RS strategy lead to improved final HSI classification performance.

Importantly, SegEarth-OV does not directly use the final CLIP patch outputs for dense matching. Instead, to mitigate the mismatch caused by directly upsampling the final projected tokens, it extracts an intermediate representation from the input tokens of the last Transformer block. Let the patch size be P (e.g., $P = 16$), so that $h = H/P$ and $w = W/P$. Denote by

$$\mathbf{X}^{(L)} = \left[\mathbf{x}_{\text{cls}}^{(L)}, \mathbf{x}_1^{(L)}, \ldots, \mathbf{x}_{hw}^{(L)}\right]^\top \in \mathbb{R}^{(hw+1)\times d} \tag{1}$$

the token sequence fed into the last Transformer block (block index L), where $\mathbf{x}_{\text{cls}}^{(L)}$ is the global [CLS] token, and $\mathbf{x}_i^{(L)}$

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is patch tokens. SegEarth-OV then applies the original CLIP projection head Proj(·) to the patch tokens of this intermediate layer to obtain low-resolution dense features

$$\mathbf{F}_{\mathrm{LR}} = \mathrm{Proj}\left(\mathbf{X}_{\mathrm{patch}}^{(L)}\right) \in \mathbb{R}^{(hw)\times c}$$
$$\mathbf{X}_{\mathrm{patch}}^{(L)} = \left[\mathbf{x}_1^{(L)}, \ldots, \mathbf{x}_{hw}^{(L)}\right]^{\top}. \tag{2}$$

To obtain pixel-aligned representations, SegEarth-OV further employs SimFeatUp [25] to upsample and refine $\mathbf{F}_{\mathrm{LR}}$ to the original spatial resolution

$$\mathbf{F}_{\mathrm{HR}} = \mathrm{SimFeatUp}\left(\mathbf{F}_{\mathrm{LR}}, \mathbf{I}\right) \in \mathbb{R}^{H\times W\times c} \tag{3}$$

where SimFeatUp is a lightweight, once-trained feature upsampler guided by the high-resolution RGB image $\mathbf{I}$, and is used as a plug-and-play module during inference. Meanwhile, CLIP-T encodes each category name into a normalized text embedding $\mathbf{t}_k \in \mathbb{R}^c$, $k = 1, \ldots, K$, forming $\mathbf{T} = [\mathbf{t}_1, \ldots, \mathbf{t}_K]^{\top} \in \mathbb{R}^{K\times c}$. After obtaining the pixel-wise visual feature $\mathbf{f}(p) \in \mathbb{R}^c$ at pixel p from $\mathbf{F}_{\mathrm{HR}}$, we compute open-vocabulary classification scores via cosine similarity

$$s_{p,k} = \cos\left(\mathbf{f}\left(p\right), \mathbf{t}_k\right) = \frac{\mathbf{f}\left(p\right)^{\top}\mathbf{t}_k}{\|\mathbf{f}\left(p\right)\|_2 \ \|\mathbf{t}_k\|_2}. \tag{4}$$

Following SegEarth-OV, we suppress the CLIP "global bias" by subtracting a scaled global embedding from each pixel feature:

$$\widetilde{\mathbf{f}}\left(p\right) = \mathbf{f}\left(p\right) - \lambda\, \mathbf{o}_{\mathrm{cls}} \tag{5}$$

$$s_{p,k} = \cos\left(\widetilde{\mathbf{f}}\left(p\right), \mathbf{t}_k\right) = \frac{\widetilde{\mathbf{f}}\left(p\right)^{\top}\mathbf{t}_k}{\left\|\widetilde{\mathbf{f}}\left(p\right)\right\|_2 \ \|\mathbf{t}_k\|_2} \tag{6}$$

where $\mathbf{o}_{\mathrm{cls}}$ is the projected `[CLS]` embedding, and λ is a fixed scaling factor. Finally, we normalize the scores over all categories (with temperature τ) to obtain pixel-wise class probabilities

$$p_{p,k} = \frac{\exp\left(s_{p,k}/\tau\right)}{\sum_{k'=1}^{K}\exp\left(s_{p,k'}/\tau\right)}. \tag{7}$$

In this way, we obtain a probability distribution for every pixel, which can be directly used for PLG and subsequent refinement.

The above equations describe how the selected SegEarth-OV module produces dense open-vocabulary probabilities. They are included for completeness and reproducibility rather than to claim the SegEarth-OV operations themselves as new. In our PLG stage, the HSI-specific adaptations are: 1) physically aligned spectral interpolation from HSI to an RGB-compatible proxy image; 2) confidence-aware pseudo-label construction using BvSB margins; and 3) multiresolution prediction fusion through the proposed RS strategy. These adaptations make the pseudo-labels more suitable for the subsequent spectral learner, which operates on the full hyperspectral cube rather than on the RGB proxy.

In order to obtain not only a hard pseudo-label (i.e., the most likely category) but also a measure of its reliability, we introduce a confidence measure based on the BvSB criterion [42], [43]. Formally, let

$$p_i = (p_{i1}, p_{i2}, \ldots, p_{iC})^{\mathrm{T}}$$

be the classification probability vector of the ith pixel x_i, where C denotes the total number of categories, and p_{ij} represents the probability that pixel x_i belongs to class j. We then define the BvSB-based confidence score for pixel x_i as

$$\mathrm{BvSB}\left(x_i\right) = P_B\left(p_i\right) - P_{SB}\left(p_i\right) \tag{8}$$

where $P_B(p_i)$ denotes the highest probability value in p_i (corresponding to the predicted class), and $P_{SB}(p_i)$ denotes the second highest probability value in p_i. By definition, the BvSB score is non-negative and reflects the margin between the most likely and the second most likely classes. A larger BvSB value indicates that the model clearly prefers one class over all others, which means that the corresponding prediction is more reliable and less ambiguous. In contrast, when the difference between the top two probabilities is small, the model is uncertain: the pixel may lie near a decision boundary, correspond to a mixed region, or simply be misclassified. Therefore, a low BvSB value suggests that the pseudo-label assigned to this pixel should be treated carefully. Based on these confidence scores, we divide the pseudo-labeled pixels into several confidence levels by setting appropriate thresholds on the BvSB values. Pixels with high BvSB are grouped into confident subsets, while those with medium or low BvSB form hard subsets, respectively. This confidence-aware partition is important in the subsequent noise-robust learning (NLL) stage. It provides a clear way to construct clean, confident, and hard sets from the pseudo-labels. Clean and highly confident pixels can be used with strong supervision signals, which helps stabilize the optimization process, whereas hard or low-confidence pixels are used with specially designed supervisory strategies. In this manner, our framework uses CLIP-based PLG not only to obtain semantic annotations but also to estimate their reliability, which supports more robust training with noisy pseudo-labels.

However, unlike images in natural scenes, remotely sensed HSIs often have very large spatial dimensions, sometimes covering entire cities or regions with very high resolutions. Directly feeding such large images into CLIP-based models is computationally expensive and requires a large amount of memory. A commonly used strategy to reduce this cost is the *patchify* operation, in which the large-scale imagery is divided into a set of smaller, possibly overlapping patches using a sliding window. Each patch is processed independently by the model to obtain a local prediction map, and the final prediction for the whole image is reconstructed by combining the predictions of all patches.

Although this patch-based strategy is practical, it has several limitations in the context of remote sensing segmentation. Because the receptive field of CLIP is fixed and object scales in remote sensing scenes vary widely, the model may fail to capture both global context and fine local details at the same time. If the patch size is small, large-scale structures (e.g., long roads or large parking lots) may be fragmented or only partially visible within each patch, which leads to incomplete or inconsistent predictions. On the other hand, if the patch size is large, small objects (e.g., cars, narrow alleys, or thin roads) may occupy only a small region of the patch and thus be dominated by the surrounding background, causing them to be

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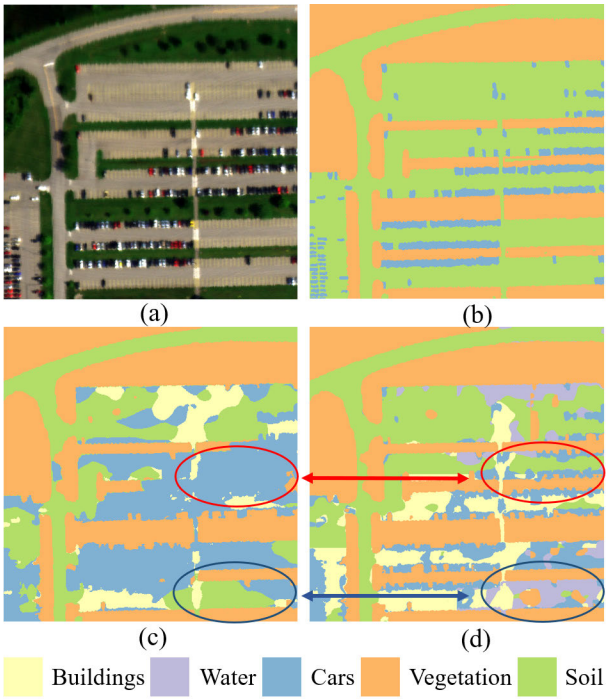


Fig. 2. Comparison of prediction results at different image scales. (a) Original image. (b) Ground truth. (c) Prediction result without any image upsampling, where the model directly operates at the original resolution. (d) Prediction result with 2× image upsampling before inference. The comparison shows that higher input resolution improves the detection of small objects such as individual cars and thin structures, while a lower resolution is more suitable for capturing large homogeneous areas such as continuous road regions. This indicates that different image scales emphasize complementary aspects of the scene.

missed or misclassified. In addition, patch-wise processing can introduce artifacts at patch boundaries, which further harms the spatial consistency of the final prediction map.

The influence of input scale on CLIP-based segmentation is illustrated in Fig. 2. When the image resolution is relatively low, CLIP models tend to capture the global layout and identify large semantic regions such as parking lots and continuous roads, but they may miss local details like thin or discontinuous road segments. In contrast, when the image is upsampled to a higher resolution, the model can more easily recognize small objects such as cars and other fine structures, but large objects such as long roads may not appear at a suitable scale, which may cause them to be fragmented or incorrectly classified. This tradeoff shows that a single resolution is not sufficient to achieve good performance on both small and large structures in complex remote sensing scenes.

To address these problems and better use the complementary information at different image scales, we propose a simple but effective RS strategy that fuses prediction results obtained at multiple resolutions. Given an input image $\mathbf{I}$, we first generate several scaled versions of $\mathbf{I}$ by applying different upsampling factors $\{s_1, s_2, \ldots, s_N\}$. For each scale s_i, we upsample the image using an upsampling operator $\mathbf{U}_{s_i}$, feed the upsampled image into the CLIP-based segmentation pipeline to obtain a probability map at that scale, and then map the probability map back to the original resolution using a downsampling operator $\mathbf{D}_{s_i}$. The final prediction $\mathbf{P}$ is obtained by averaging the probability maps across all scales

$$\mathbf{P} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{D}_{s_i}\left(\mathrm{CLIP}\left(\mathbf{U}_{s_i}(\mathbf{I})\right)\right) \tag{9}$$

where s_i denotes the upsampling factor at the ith scale, and $\mathbf{D}$ and $\mathbf{U}$ represent the downsampling and upsampling operations, respectively.

In our work, both upsampling and downsampling operations are implemented using bicubic interpolation, which offers a good balance between computational efficiency and interpolation quality. Bicubic interpolation is widely used in image processing and usually introduces few artifacts, which makes it suitable for our multiscale fusion strategy. In principle, more advanced single-image super-resolution methods or learned upsampling modules could replace the bicubic interpolation operators $\mathbf{U}_{s_i}$ and $\mathbf{D}_{s_i}$ and may further improve performance, especially for very small objects. However, our experiments show that even this simple choice of interpolation already brings clear improvements.

B. Noisy Label Learning

In this stage, we explicitly incorporate spectral information to further improve the classification accuracy obtained from the previous PLG stage. The overall training process consists of two phases: a *warmup phase* and a *label refinement phase*. Compared with the RGB proxy used in the PLG stage, the original hyperspectral measurements contain rich material-specific spectral signatures that are more stable across changes in illumination, season, and atmospheric conditions. By directly modeling these spectral distributions, the NLL stage can correct systematic biases inherited from CLIP-based pseudo-labels and adapt the decision boundaries to the statistical properties of the target scene.

More concretely, the pseudo-labels produced by SegEarth-OV are generated in the RGB space and inevitably suffer from domain gaps between natural images and remote sensing data. These domain gaps lead to label noise, especially for confusing categories or boundary regions. In contrast, hyperspectral pixels from different classes often exhibit distinct spectral curves, which provide complementary information that is not fully exploited in the PLG stage. Therefore, our NLL stage aims to transfer the coarse semantic prior from the RGB-based pseudo-labels into a purely spectral model, and then refine these labels by exploiting the underlying spectral distributions.

The division of the NLL stage into a warmup phase and a label refinement phase is logically necessary. The GMM-based refinement depends on meaningful predictions, reliable BvSB margins, and partially organized spectral feature distributions. At the very beginning of training, however, the spectral classifier is randomly initialized, so its predictions and confidence scores are not informative enough for density modeling. The warmup phase therefore first distills the coarse semantic prior from SegEarth-OV into the HSI classifier using class-balanced, confidence-weighted pseudo-label sampling. Only after this initialization can the label refinement phase safely separate confident and hard samples and fit class-wise

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spectral distributions. This sequential design reduces early confirmation bias and prevents the GMM from modeling random network outputs.

1) Warmup Phase: In the first warmup training phase, we use the pseudo-labels provided by SegEarth-OV to train an HSI classification network, namely, MambaHSI [18], with the standard cross-entropy loss. MambaHSI is a recent hyperspectral classification model that directly operates on spectral vectors and can capture long-range dependencies along the spectral dimension. Since the input of MambaHSI is the original hyperspectral data rather than the interpolated RGB image, this warmup training encourages the model to learn a mapping from spectral signatures to the semantic categories implied by CLIP-based pseudo-labels.

Due to the extreme class imbalance and significant label noise present in remote sensing scenes, directly training on all pseudo-labeled samples may cause the model to be dominated by large classes and overfit to noisy labels. To avoid this, we adopt a class-balanced sampling strategy: in each training iteration, a fixed number of samples is randomly drawn from each class, instead of using all available labeled pixels. The resulting mini-batch, together with its pseudo-labels, is referred to as the *random set*. This strategy ensures that each class is observed with a similar frequency during training, increases the diversity of the training samples, and prevents the model from relying too heavily on a few dominant categories.

Furthermore, we make use of the BvSB values provided by SegEarth-OV as sampling weights. Samples with higher BvSB scores, which correspond to more confident predictions, are more likely to be selected, while those with low confidence are sampled less frequently. Each class is sampled independently, so that both rare classes and common classes benefit from a focus on reliable pseudo-labels. Overall, this warmup process can be regarded as distilling the coarse semantic prior from the PLG stage into a spectral model that no longer depends on the RGB proxy.

2) Label Refinement Phase: After the warmup phase, the spectral classifier already captures useful information from the noisy pseudo-labels. However, the remaining label noise still limits the final performance. To further reduce the influence of noisy labels and exploit the spectral structure of the data, we introduce a label refinement phase. In this phase, we construct two additional training sets, namely, the *confident set* and the *hard set*, and assign soft labels to their samples based on a class-wise GMM. The overall procedure consists of the following four steps.

1) For the region of interest, we first use the warmup-trained MambaHSI network to obtain predicted labels and the corresponding BvSB values for all pixels. For each class, we collect the predicted samples belonging to that class and apply a GMM to the distribution of their BvSB values. The GMM typically contains two components, which naturally divide the samples into a high-confidence group and a low-confidence group. The spectral samples associated with the high-confidence group are termed the *confident set*, while those associated with the low-confidence group form the *hard set*. The confident set is expected to be relatively clean, containing mostly correctly labeled pixels, whereas the hard set is more informative but may contain a higher proportion of mislabeled samples. This step separates easy and hard examples in an adaptive way rather than relying on a fixed confidence threshold.
2) Next, we perform PCA on the spectral vectors to reduce their dimensionality. For each class, we then use a GMM to fit the distribution of the reduced spectral features from the confident set of that class. In this way, we obtain a class-conditional density model that describes how clean spectral samples of that class are distributed in the low-dimensional PCA space.
3) Based on the fitted GMMs, we further assign soft labels to both confident and hard samples. For each sample x (represented by its reduced spectral feature), we compute its probability density under the distribution of each class. These densities are then normalized across all classes to form a probability vector, which is used as a soft label. In other words, instead of assigning a single hard class label, we estimate how likely the sample belongs to each class according to the learned spectral distributions.
4) Finally, we include samples from the random set, confident set, and hard set into the training process. The random set preserves the semantic prior from the PLG stage, the confident set provides relatively clean supervision, and the hard set encourages the model to explore ambiguous regions and refine its decision boundaries. The generated soft labels for the confident and hard sets help to down-weight unreliable classes and provide a tradeoff between exploration (learning from difficult samples) and exploitation (trusting reliable spectral evidence).

Mathematically, the soft label $\tilde{y}(x)$ for a sample with reduced spectral feature x is calculated as

$$\tilde{y}_k(x) = \frac{\sum_{m=1}^{M} \pi_{km} \mathcal{N}(x \mid \mu_{km}, \Sigma_{km})}{\sum_{k'=1}^{K} \sum_{m'=1}^{M} \pi_{k'm'} \mathcal{N}(x \mid \mu_{k'm'}, \Sigma_{k'm'})} \tag{10}$$

where K is the number of classes, M is the number of Gaussian components per class, and π_{km}, μ_{km}, and Σ_{km} represent the weight, mean, and covariance of the mth Gaussian component of the kth class, respectively. The term $\mathcal{N}(x \mid \mu_{km}, \Sigma_{km})$ denotes the value of the Gaussian density function at x. The kth component $\tilde{y}_k(x)$ is thus the normalized class-conditional density and can be interpreted as the probability that x belongs to class k under the GMM model. This construction naturally produces softer labels for samples located in overlapping regions and sharper labels for samples lying in high-density regions of a particular class.

This density-based soft labeling mitigates pseudo-label noise in two complementary ways. First, for samples located near a reliable class-specific spectral cluster, the normalized density is dominated by that class, so the model receives a strong but data-adaptive supervision signal. Second, for ambiguous or potentially mislabeled samples that lie between several class distributions, the posterior probabilities are naturally softened instead of being forced to follow an incorrect one-hot pseudo-label. Therefore, the GMM does not simply

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remove samples; it converts the spectral distribution information into probabilistic supervision, which down-weights unreliable classes and reduces the risk of overfitting to noisy CLIP labels.

The total training loss of the NLL stage is defined as

$$L = \mathrm{CE}(\hat{y}_r, \tilde{y}_r) + \lambda_1 \mathrm{CE}(\hat{y}_c, \tilde{y}_c) + \lambda_2 \mathrm{CE}(\hat{y}_h, \tilde{y}_h) \tag{11}$$

where CE denotes the cross-entropy loss, $\hat{y}_r$, $\hat{y}_c$, and $\hat{y}_h$ are the predicted class probability vectors for the random, confident, and hard sets, respectively, $\tilde{y}_r$ is the pseudo-label provided by SegEarth-OV for the random set, and $\tilde{y}_c$ and $\tilde{y}_h$ are the soft labels generated by the GMM for the confident and hard sets. The scalars λ_1 and λ_2 are hyperparameters that control the relative contributions of the confident and hard sets to the overall objective. In this loss, the first term encourages the model to be consistent with the original CLIP-based pseudo-labels and maintains the semantic prior learned from the PLG stage. The second term exploits the relatively clean confident set with soft labels, which helps to refine the decision boundaries using reliable spectral evidence. The third term allows the model to benefit from the informative but noisy hard set, while the soft labels and weighting factor λ_2 reduce the risk of overfitting to incorrect labels. Together, these three terms enable the model to gradually correct noisy labels, sharpen the class boundaries in the spectral space, and achieve robust classification performance under severe label noise.

Overall, the discriminative spectral information is recovered precisely because of the architectural decoupling between Stage 1 and Stage 2. In Stage 1, the RGB image acts strictly as a "proxy" to bridge the modality gap, allowing the natural-image-trained CLIP to generate initial semantic pseudo-labels. However, in Stage 2, the MambaHSI network takes the full, original hyperspectral data as its input, guided by the pseudo-labels from Stage 1. Because pixels of the same land-cover class inherently share similar high-dimensional spectral signatures, the MambaHSI network relies on this spatial–spectral homogeneity to correct the pseudo-label noise.

IV. Experimental Results

A. Dataset

We conducted experiments on three publicly available datasets, including Chikusei [49], AeroRIT [50], and Pavia Center [51]. A brief introduction is presented as follows.

1) Pavia Center: Pavia Center scene is a widely used airborne urban hyperspectral benchmark acquired by the ROSIS sensor during a flight campaign over Pavia in northern Italy. It contains 102 spectral bands covering the spectral range of 430–860 nm, with a size of 1095 × 751. The geometric resolution is 1.3 m. There are nine land cover categories in the dataset, including water, trees, meadows, bricks, bare soil, asphalt, bitumen, tile, and shadows. Owing to the coexistence of natural surfaces, man-made materials, and shaded regions, this dataset is particularly suitable for evaluating fine-grained urban hyperspectral classification methods.

2) AeroRIT: The AeroRIT dataset was captured with a visible near-infrared (VNIR) hyperspectral Headwall Photonics Micro Hyperspec E-Series CMOS sensor. It contains 372 spectral bands covering the spectral range of 397–1003 nm. The ground sampling distance is about 0.4 m. For reproducibility, the retained spatial region starts at row 500 and column 100 of the original image, with a spatial size of 1024 × 3072 pixels. This fixed crop is selected from the main valid annotated area of the AeroRIT scene and is applied identically to all compared methods before evaluation. In addition, we removed the first and last ten spectral bands due to significant data noise in these bands. As illustrated in Fig. 3, the global mean spectrum remains relatively stable over the central wavelength range, but exhibits abnormal behavior at both spectral boundaries. Keeping these noisy boundary bands may negatively affect the hyperspectral classifier and may also introduce unstable spectral statistics during noisy-label learning. Therefore, removing the first and last ten bands is a sensor-noise-driven preprocessing step rather than a method-specific tuning strategy. The same processed spectral cube is used for all methods, and no label information or classification result is used to determine which bands are removed. There are five distinct land cover categories in the dataset, including buildings, water, cars, vegetation, and soil.

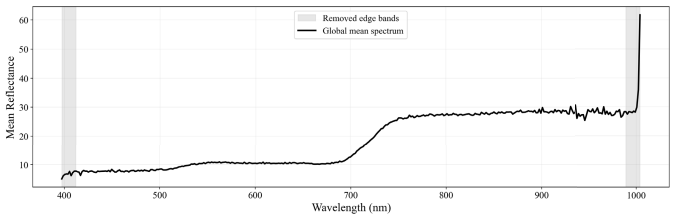


Fig. 3. Global mean spectrum across all spectral bands of the AeroRIT dataset, where the shaded regions indicate the removed boundary bands.

3) Chikusei: The dataset was captured using a Headwall Nano-Hyperspec sensor. It contains 128 spectral bands covering the spectral range of 343–1018 nm, and the ground sampling distance is 2.5 m. For category consolidation, the purpose is to construct a semantically clear land-cover taxonomy that is compatible with zero-shot open-vocabulary classification. We consolidated highly similar classes within the dataset labels, such as "bare soil (farmland)" and "bare soil (park)," resulting in seven distinct land cover classes, including farmland, bare soil, water, road, grass, trees, and buildings. These fine-grained labels describe very similar materials or land-cover semantics and are difficult to distinguish reliably using generic text prompts in a zero-shot setting. The class mapping is fixed before evaluation and applied identically to all methods, so the consolidation does not introduce a method-specific advantage. For reproducibility, the retained region starts at row 100 and column 100 of the original image, with a spatial size of 2048 × 2048 pixels. This deterministic crop is selected from the main valid region of the scene and is fixed before running any method. Therefore, all compared methods are evaluated on exactly the same spatial region and the same ground-truth mask. This is a preprocessing choice intended to exclude potentially invalid or weakly informative border regions, such as boundary pixels with incomplete spatial support or limited annotation reliability. The resulting square subset also facilitates patch-based training and evaluation. We leave finer-grained open-vocabulary HSI classification for

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TABLE I
NUMBER OF LABELED PIXELS IN DIFFERENT DATASETS

Chikusei		Pavia Centre		AeroRIT	
Category	Total	Category	Total	Category	Total
Farmland	16289	Water	65971	Buildings	536188
Soil	7458	Trees	7598	Water	46480
Road	801	Meadows	3090	Cars	91268
Grass	11852	Bricks	2685	Vegetation	1474345
Trees	20516	Bare Soil	6584	Soil	993792
Buildings	3604	Asphalt	9248		
Water	2586	Bitumen	7287		
		Tile	42826		
		Shadows	2863		

future work; for example, reasoning-capable multimodal large language models could be introduced to exploit scene-level semantics and land-use context for distinguishing visually and spectrally similar subtypes.

Table I summarizes the number of labeled pixels for each class in the three datasets. It can be observed that there exists a significant class imbalance, which poses challenges for model training. Given the considerable impact of prompt engineering on CLIP-based pseudo-labeling [26], we deliberately designed prompts that are semantically refined rather than strictly aligned with the category names. Specifically, when generating pseudo-labels using CLIP, we replaced "water" with "river or lake" as the prompt. This is because CLIP was trained on natural images, and using "water" directly could cause confusion, often misclassifying blue-colored buildings as water bodies. In addition, we replaced "vegetation" with "grass or trees," which provides a more specific and meaningful description, improving the quality of the generated pseudo-labels.

B. Implementation Details and Evaluation Metrics

The hyperspectral classifier (i.e., MambaHSI) was trained for 10, 15, and 20 epochs on AeroRIT, Chikusei, and Pavia Center, respectively, with 20 iterations per epoch. The initial learning rate was set to $1e^{-3}$ for AeroRIT, $2e^{-3}$ for Chikusei, and $4e^{-4}$ for Pavia Center, and decayed using cosine annealing, with the minimum learning rate $\eta_{\min}$ set to $1e^{-5}$ for Chikusei and AeroRIT, and $1e^{-4}$ for Pavia Center. In the first half of training, the model was trained with pseudo-labels generated by CLIP (i.e., SegEarth-OV). After half of the training iterations, the confident set and hard set were incorporated for training. In our work, λ_1 is set to 0.2, and λ_2 is set to 1. We adopted Adam as the optimizer. When generating pseudo-labels using SegEarth-OV, we adopted a window size of 224×224 and a stride of 112 for the Chikusei and Pavia Center datasets, and a window size of 448×448 with a stride of 224 for the AeroRIT dataset due to higher ground resolution. For the spectral interpolation method, we utilize linear interpolation along the spectral dimension to synthesize the target bands. For a given target wavelength, we map it to a fractional index t within the original HSI wavelength library. We identify the two nearest adjacent spectral bands—the floor index $\lfloor t \rfloor$ and the ceiling index $\lceil t \rceil$—and assign them linear weights $w_1 = 1 - (t \bmod 1)$ and $w_2 = t \bmod 1$, respectively. A transformation matrix is constructed from these weights to generate the interpolated band via a weighted sum. Furthermore, the specific target wavelengths (655, 553, and 451 nm) were chosen because they precisely align with the physical center wavelengths of the RGB bands in standard optical imaging. This selection minimizes the domain gap between our HSI data and the natural RGB images that the CLIP model was originally trained on. For the RS proposed in Section III-A, we fused the prediction results from $1\times$ and $2\times$ upsampled images. Regarding the factor λ and temperature τ in the PLG stage, following the default configuration of SegEarth-OV, we set the factor λ to -0.3 and the temperature τ to 1 for PLG. This ensures a fair comparison and preserves the transfer of open-vocabulary semantic priors from the VLM without dataset-specific bias. To enhance the generalization ability of the network and prevent overfitting, we added Gaussian noise with a standard deviation of 0.1 to the HSI input during training. Three commonly used evaluation metrics, including overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ), are adopted to evaluate the classification performance.

C. Comparison With SOTAs

We compare our method with several state-of-the-art CLIP-based classification methods, including CLIP [20], MaskCLIP [38], SCLIP [39], GEM [40], ClearCLIP [41], and SegEarth-OV [25]. For a fair comparison, these approaches are evaluated under the same zero-shot setting and on the same three benchmark datasets, namely, Pavia Center, AeroRIT, and Chikusei, using identical class definitions and prompts. In addition, we extended our comparisons by including five recent HSI classification methods, namely, CTF-SSCL [44], DSNet [45], HyperTaFOR [46], MCTGCL [47], and Semi-HCN [48]. These hyperspectral methods are trained in a one-shot setting, i.e., each class is provided with one labeled training sample. In contrast, our method operates in a zero-shot manner and does not use any labeled samples from the target dataset. Therefore, the compared methods actually rely on more task-specific supervision than our approach. Despite this advantage, their performance is still consistently inferior to ours on the evaluated datasets. The quantitative results are summarized in Tables II–IV, while representative qualitative results are shown in Figs. 4–6. These visualizations help to intuitively assess the spatial coherence of the predicted maps and the capability of each method in capturing object boundaries and small structures.

As can be seen from the tables, our method achieves strong and stable performance in terms of OA, AA, and kappa coefficient (κ) on all three datasets. In particular, compared with the CLIP-based dense prediction baselines, our framework substantially improves the classification metrics by further exploiting hyperspectral measurements and reducing pseudo-label noise. On the AeroRIT dataset, which contains complex urban structures and small objects such as cars, our method exhibits a clear performance gain, reflecting its improved ability to handle fine-scale details and strong interclass confusion. A similar trend can be observed on the Chikusei

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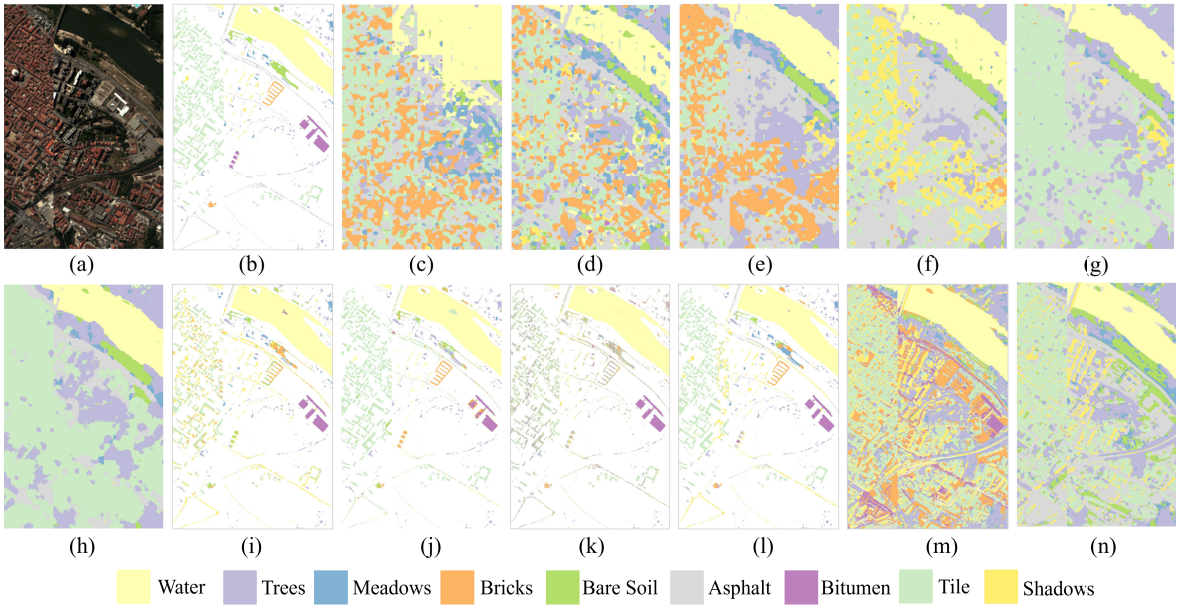


Fig. 4. Visualization of the classification maps produced by different approaches on Pavia Center dataset. (a) False-color image derived from the hyperspectral data. (b) Ground truth reference map. (c) CLIP. (d) MaskCLIP. (e) SCLIP. (f) GEM. (g) ClearCLIP. (h) SegEarth-OV. (i) CTF-SSCL. (j) DSNet. (k) HyperTaFOR. (l) MCTGCL. (m) Semi-HCN. (n) Proposed method (ours). The comparison shows that our method generates cleaner and more coherent segmentation maps, preserves object boundaries better, and reduces misclassifications in complex urban regions compared with existing baselines. Note: (i)–(l) are patch-wise methods, so only results in GT-annotated regions are shown.

TABLE II

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON PAVIA CENTER DATASET. THE BEST CLASS-SPECIFIC, OA, AA, AND κ VALUES ARE IN BOLD

Method	Water	Trees	Meadows	Bricks	Bare Soil	Asphalt	Bitumen	Tile	Shadows	OA(%)	AA(%)	κ(%)
CLIP [20]	83.73	1.42	0.32	12.22	0.00	8.46	0.00	53.86	0.45	53.69	17.83	35.29
MaskCLIP [38]	91.71	83.25	**77.38**	1.45	66.40	85.22	0.00	60.48	4.51	72.59	52.27	62.90
SCLIP [39]	99.45	85.25	16.70	0.00	43.26	88.47	0.00	45.23	1.15	69.55	42.17	59.02
GEM [40]	98.35	82.80	24.17	0.00	68.92	96.84	0.00	75.32	27.17	79.95	52.62	72.21
ClearCLIP [41]	**99.97**	91.60	31.65	0.00	65.01	89.10	0.00	99.04	3.11	87.02	53.28	81.33
SegEarth-OV [25]	99.93	92.98	13.46	0.00	76.17	63.47	0.00	**99.42**	0.14	85.64	49.51	79.00
CTF-SSCL [44]	99.43	38.82	70.08	54.26	34.47	27.68	53.80	76.76	95.56	78.65	61.21	70.25
DSNet [45]	60.27	41.32	49.31	75.19	33.72	56.30	57.97	62.12	64.95	58.42	55.68	50.34
HyperTaFOR [46]	78.83	18.73	23.88	31.92	9.45	16.65	36.51	42.06	68.82	53.89	36.32	40.03
MCTGCL [47]	98.78	32.07	56.18	85.71	23.95	59.86	**62.21**	96.66	76.41	85.63	65.76	79.66
Semi-HCN [48]	99.87	68.36	53.92	**88.62**	26.04	77.21	34.12	75.32	60.21	81.30	64.85	73.95
Ours	98.70	**98.99**	34.30	0.00	**98.31**	**99.46**	0.00	98.24	**98.71**	**90.62**	**69.63**	**86.75**

TABLE III

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE AERORIT DATASET. THE BEST CLASS-SPECIFIC, OA, AA, AND κ VALUES ARE IN BOLD

Method	Buildings	Water	Cars	Vegetation	Soil	OA(%)	AA(%)	κ(%)
CLIP [20]	34.72	31.99	44.92	34.78	6.18	25.97	30.52	-1.29
MaskCLIP [38]	69.05	51.42	92.37	80.75	34.32	63.97	65.58	48.74
SCLIP [39]	67.62	26.07	88.38	93.35	20.74	64.86	59.23	46.27
GEM [40]	68.72	68.18	**92.91**	82.90	37.37	66.15	70.02	51.13
ClearCLIP [41]	72.14	33.28	90.90	88.96	38.83	69.47	64.82	54.28
SegEarth-OV [25]	75.60	20.35	70.84	94.81	33.48	70.34	59.02	54.48
CTF-SSCL [44]	50.38	76.80	25.13	62.38	40.09	52.41	50.96	33.96
DSNet [45]	35.86	55.18	13.12	55.36	33.20	43.79	38.54	26.73
HyperTaFOR [46]	47.25	87.19	7.09	31.09	20.11	30.51	38.55	8.00
MCTGCL [47]	48.62	81.90	29.09	69.09	37.38	54.60	53.22	34.95
Semi-HCN [48]	37.91	**90.02**	15.61	53.50	63.22	53.35	52.05	37.90
Ours	**89.86**	82.37	77.31	**95.98**	**64.08**	**84.11**	**81.92**	**76.51**

dataset, where our model achieves the highest OA, AA, and κ, demonstrating that it generalizes well across different sensors, spatial resolutions, and scene types.

From the qualitative comparisons in Figs. 4–6, it is clear that our method produces smoother and more coherent classification maps, with fewer isolated misclassified pixels and

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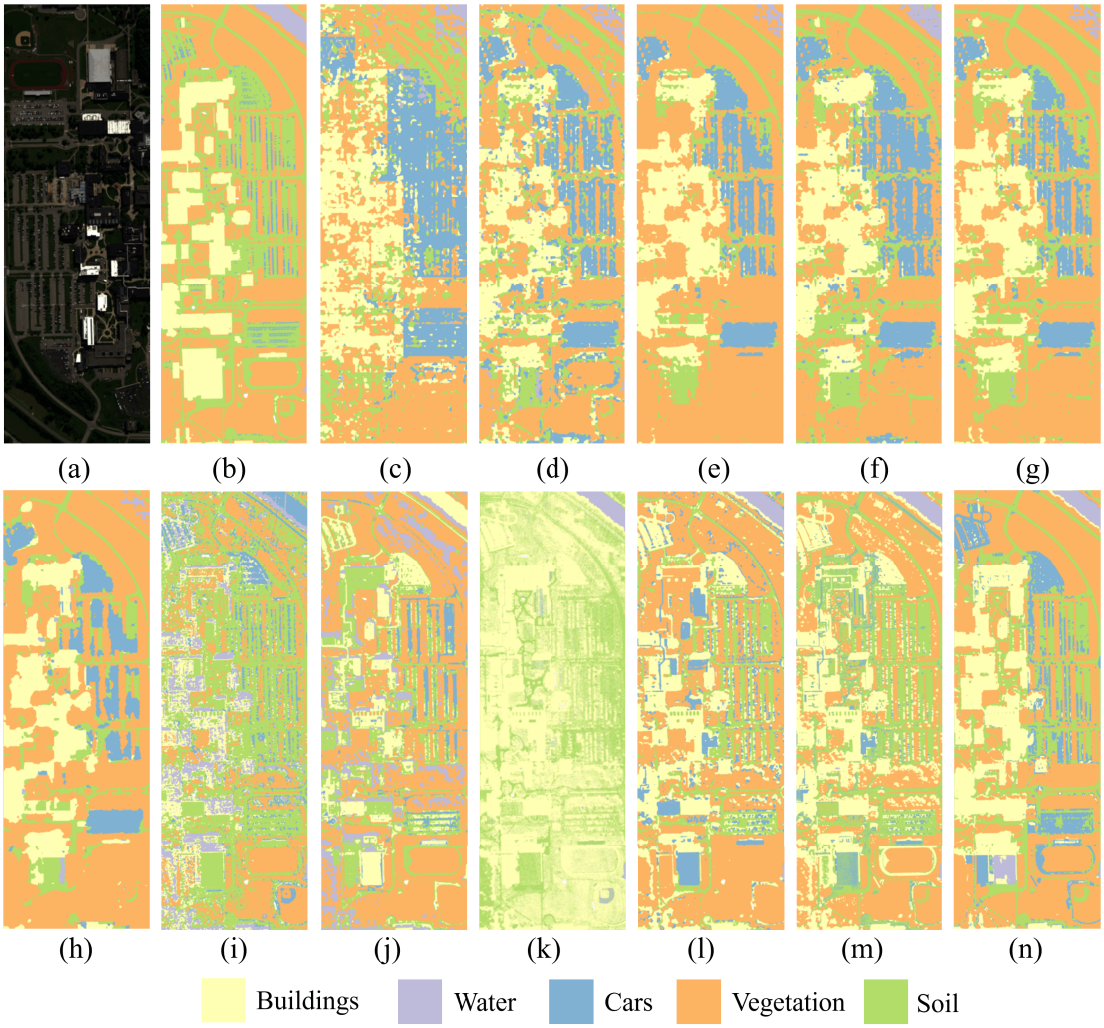


Fig. 5. Visualization of the classification maps provided by different approaches on the AeroRIT dataset. (a) False-color image derived from the hyperspectral data. (b) Ground truth reference map. (c) CLIP. (d) MaskCLIP. (e) SCLIP. (f) GEM. (g) ClearCLIP. (h) SegEarth-OV. (i) CTF-SSCL. (j) DSNet. (k) HyperTaFOR. (l) MCTGCL. (m) Semi-HCN. (n) Proposed method (ours). Note: (i)–(l) are patch-wise methods, so only results in GT-annotated regions are shown.

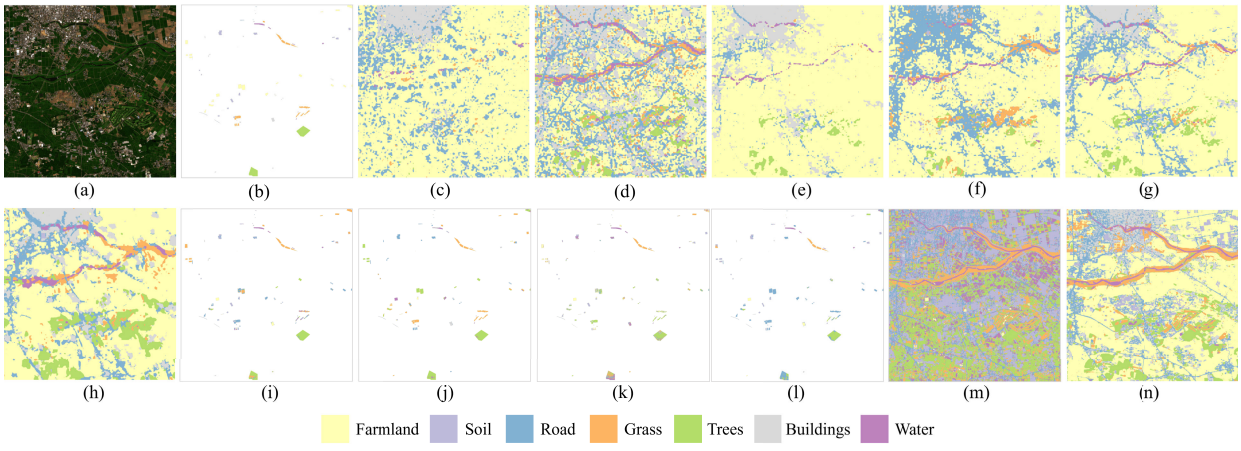


Fig. 6. Visualization of the classification maps provided by different approaches on the Chikusei dataset. (a) False-color image derived from the hyperspectral data. (b) Ground truth reference map. (c) CLIP. (d) MaskCLIP. (e) SCLIP. (f) GEM. (g) ClearCLIP. (h) SegEarth-OV. (i) CTF-SSCL. (j) DSNet. (k) HyperTaFOR. (l) MCTGCL. (m) Semi-HCN. (n) Proposed method (ours). Note: (i)–(l) are patch-wise methods, so only results in GT-annotated regions are shown.

clearer region boundaries. Competing CLIP-based methods often suffer from noisy predictions and fragmented regions, especially in areas where the CLIP features at low resolution are not sufficient to distinguish visually similar classes. In

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TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE CHIKUSEI DATASET. THE BEST CLASS-SPECIFIC, OA, AA, AND κ VALUES ARE IN BOLD

Method	Farmland	Soil	Road	Grass	Trees	Buildings	Water	OA(%)	AA(%)	κ(%)
CLIP [20]	60.06	0.00	0.00	0.00	0.00	2.72	0.00	15.66	8.97	-4.84
MaskCLIP [38]	66.95	18.13	66.54	71.46	86.68	17.65	78.58	66.10	58.00	56.50
SCLIP [39]	**96.80**	0.00	7.87	3.58	77.19	17.59	60.75	54.35	37.68	38.49
GEM [40]	92.79	20.07	70.41	40.15	57.99	8.44	62.95	56.67	50.40	43.78
ClearCLIP [41]	94.92	10.58	37.58	27.23	83.24	12.60	72.39	62.09	48.36	49.54
SegEarth-OV [25]	87.34	26.66	43.70	66.44	99.76	28.08	67.75	75.54	59.96	67.49
CTF-SSCL [44]	30.50	67.23	56.15	35.25	82.00	22.01	75.17	54.15	52.62	43.24
DSNet [45]	14.28	31.04	45.99	**94.33**	58.38	12.96	68.37	48.18	46.48	35.46
HyperTaFOR [46]	20.75	40.26	25.44	40.02	55.12	10.77	67.18	39.24	37.08	25.32
MCTGCL [47]	26.94	62.86	75.56	35.04	77.35	23.41	74.14	51.44	53.61	40.19
Semi-HCN [48]	24.14	64.88	78.80	64.30	88.29	12.16	93.29	60.19	60.84	49.58
Ours	74.52	**98.85**	**84.64**	90.78	**99.99**	**47.00**	**100.00**	**88.33**	**85.11**	**84.98**

contrast, our results show more homogeneous regions for large land-cover types (e.g., meadows, farmland, vegetation) and better preservation of narrow or small structures such as roads and cars. These observations are consistent with the quantitative improvements in AA, which measures the average performance across classes and is sensitive to the accuracy on rare or difficult categories.

The performance improvement can mainly be attributed to three aspects of our framework: 1) the incorporation of spectral information in the NLL stage; 2) the RS strategy in the PLG stage; and 3) the label refinement strategy based on GMMs. First, instead of relying solely on RGB proxies, our method explicitly uses full hyperspectral measurements in the NLL stage. Hyperspectral pixels from the same class often share similar spectral signatures, and this spectral homogeneity naturally reduces the impact of mislabeled samples. Even when the pseudo-labels from CLIP contain noticeable errors, the spectral classifier can still learn robust decision boundaries by exploiting the relationships between spectrally close samples. Second, the RS strategy fuses predictions from multiple resolutions, which helps to balance global context and local detail: low-resolution inputs emphasize large-scale structures and smooth region shapes, while high-resolution inputs highlight small objects and thin structures. By averaging predictions from different scales, we obtain more stable and accurate pseudo-labels, which provide a better starting point for subsequent spectral learning. Third, the label refinement strategy uses class-wise GMMs and BvSB-based confidence to construct random, confident, and hard sets, and assigns soft labels to the latter two. This procedure allows the model to take advantage of informative but noisy samples while reducing the risk of overfitting to incorrect labels. A typical example that highlights the benefit of our design is the *water* class on the Chikusei and AeroRIT datasets. In these scenes, CLIP-provided pseudo-labels tend to misclassify many water-body pixels, often confusing them with dark roofs, shadows, or other spectrally different but visually similar regions in the RGB representation. Our NLL stage, however, can effectively learn from a small number of highly confident water samples and then generalize to other water regions by leveraging the distinctive spectral profile of water. As a result, our method achieves a significant accuracy gain on the water class compared with all CLIP-based baselines. This improvement is clearly reflected in both the class-specific accuracies and the visual results, where water bodies are more completely and cleanly delineated. In contrast, existing CLIP-based classification methods operate directly on upsampled CLIP features and struggle to recover fine-grained pixel-level semantics from low-resolution feature maps. Since the spatial resolution of CLIP features is relatively low, information about small or thin structures is often lost during feature extraction. When these features are simply upsampled, the resulting dense predictions tend to be over-smoothed and fail to capture subtle boundaries. Moreover, these methods do not explicitly exploit hyperspectral information, and thus cannot use spectral similarities to correct misclassifications introduced by noisy pseudo-labels. This limitation explains why they show inferior performance, particularly in challenging categories and complex urban scenes.

It is also important to note that, although our model significantly improves overall classification performance, it still struggles to accurately recognize certain difficult classes, such as *brick* and *bitumen* in Pavia Center dataset, and *buildings* in the Chikusei dataset. The main reasons are as follows. First, classes such as *brick* and *bitumen* occupy a small proportion of the pixels, which makes them underrepresented during training. Second, their visual and spectral characteristics are close to those of other classes, leading to ambiguous semantic information and strong confusion even for human observers. Furthermore, classes like *buildings* in the Chikusei scene are visually heterogeneous and are frequently mixed with surrounding categories such as soil, road, and boundary pixels. Third, the underlying CLIP model (SegEarth-OV), on which our pseudo-labels are based, already fails to reliably distinguish these classes at the feature level. CLIP-based dense prediction relies on relatively low-resolution visual features followed by upsampling. This mechanism tends to over-smooth boundaries and is highly ineffective for fragmented or small man-made structures. As a consequence, the pseudo-labels for brick and bitumen are highly noisy, and this noise is difficult to correct completely even with our spectral refinement strategy. For example, on the Chikusei dataset, all zero-shot CLIP-based baselines perform poorly on the *buildings* category, with class accuracies ranging from 2.72%

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TABLE V
COMPARISON WITH RECENT ZERO-SHOT HSI CLASSIFICATION METHODS

Dataset	Method	OA(%)	AA(%)	κ(%)
Pavia Centre	HZSCM [52]	**90.69**	66.17	86.66
	SpectralZero [53]	90.54	**71.79**	86.61
	Ours	90.62	69.63	**86.75**
Chikusei	HZSCM [52]	85.27	80.66	80.87
	SpectralZero [53]	83.49	81.15	78.79
	Ours	**88.33**	**85.11**	**84.98**
AeroRIT	HZSCM [52]	76.31	75.46	65.72
	SpectralZero [53]	81.71	77.53	72.80
	Ours	**84.11**	**81.92**	**76.51**

to 28.08% (e.g., CLIP, MaskCLIP, SCLIP, GEM, ClearCLIP, and SegEarth-OV as shown in Table IV), while our method substantially improves this accuracy to 47.00%, indicating that the performance bottleneck originates primarily from the CLIP model rather than our framework. We leave these limitations for future work. On one hand, future research could focus on collecting large-scale hyperspectral datasets to train an open-vocabulary multimodal model specifically designed for hyperspectral imagery. On the other hand, a hybrid approach could be adopted where the specifically challenging classes are manually annotated, while CLIP is utilized as a powerful auxiliary tool to handle the broader, visually distinct categories.

Despite these remaining challenges, our method still shows a clear advantage in terms of both OA and visual quality across all tested datasets. The combination of CLIP-based semantic priors, multiscale PLG, and noise-robust spectral learning forms an effective framework for zero-shot HSI classification. We believe that further improvements could be achieved by designing more discriminative prompts for ambiguous classes, incorporating additional spatial-context modeling, or replacing the current CLIP backbone with a stronger VLM. We leave these directions for future work.

In addition, since several closely related zero-shot HSI classification methods have recently appeared, we further compare SPECIAL with HZSCM [52] and SpectralZero [53]. For a fair and cautious comparison, we evaluate each compared method under two configurations: 1) the same experimental setting as SPECIAL, including the training schedule and iteration configuration; and 2) the native setting following the configuration reported in the corresponding original article. For each method and each dataset, we report the better result between these two configurations according to OA. This protocol avoids disadvantaging the compared methods due to a single implementation setting. The quantitative comparison is shown in Table V.

The results show that HZSCM and SpectralZero are strong and very recent baselines. On Pavia Center, the differences among the three methods are small, and different methods achieve the best value on different metrics. On Chikusei and AeroRIT, SPECIAL obtains the best OA, AA, and κ. These results do not imply that SPECIAL is universally better than all recent methods, but they do indicate that SPECIAL remains competitive on the evaluated datasets even against newer zero-shot HSI classification approaches.

TABLE VI
ABLATION RESULTS ON THE SPECTRAL INCORPORATION

Dataset	Modality	OA(%)	AA(%)	κ(%)
Pavia Centre	RGB	82.85	55.96	75.27
	HSI	**90.62**	**69.63**	**86.75**
Chikusei	RGB	79.95	80.54	74.09
	HSI	**88.33**	**85.11**	**84.98**
AeroRIT	RGB	75.55	74.22	64.88
	HSI	**84.11**	**81.92**	**76.51**

D. Ablation Study

In this section, we conduct a series of ablation experiments to better understand the contribution of each main component in our framework. Unless otherwise stated, all experiments follow the same settings as in the main results, including the network backbone, training schedule, and evaluation metrics.

1) Spectral Information Incorporation: We first study the impact of spectral information on land-cover classification performance. To this end, we construct a variant of our method in which the HSIs used in the NLL stage are replaced by their corresponding RGB images. In other words, we only keep three bands for each pixel, and all other settings are kept unchanged. Since the input dimensionality is reduced from a large number of bands to three channels, the number of principal components in the PCA step is also set to 3 to match the RGB format. Apart from this adjustment, the network architecture, training pipeline, PLG, and label refinement procedures remain the same.

The classification results of this RGB-based variant and our full HSI-based method are presented in Table VI. The comparison clearly shows that the use of full hyperspectral information leads to higher OA, higher AA, and a better kappa coefficient on all datasets. This is expected, since hyperspectral data provide much richer spectral information than RGB images: each material or land-cover type tends to have a characteristic spectral curve across many bands, which is not fully captured by only three broad RGB channels. As a result, the classifier trained on HSI can more easily separate classes that appear similar in RGB but differ in their spectral signatures.

In addition, the spectral similarity among samples of the same class helps to reduce the negative effect of noisy labels. When training with hyperspectral data, mislabeled samples are more easily detected and corrected by the model because they are surrounded by many spectrally similar neighbors with correct labels. For example, if a pixel from class A is mistakenly assigned to class B in the pseudo-labels, its spectral neighbors from class A still form a tight cluster in the spectral space. During training, the model learns that these neighbors share similar spectral features and should be assigned to the same class, which provides indirect evidence to correct the mislabeling. In contrast, when only RGB information is available, the differences between some classes become much smaller, and this kind of spectral redundancy is lost, making the training process more sensitive to label noise. Overall, these results confirm that explicitly using hyperspectral information is important for robust classification in the presence of noisy pseudo-labels.

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TABLE VII

ABLATION RESULTS ON THE SCALING STRATEGY

Dataset	Setting	Method	OA(%)	AA(%)	κ(%)
Chikusei	w/o RS	ClearCLIP	62.09	48.36	49.54
		SegEarth-OV	75.54	59.96	67.49
		Ours	**83.56**	**76.03**	**78.81**
	RS	ClearCLIP	72.14	57.76	62.64
		SegEarth-OV	77.02	65.08	69.49
		Ours	**88.33**	**85.11**	**84.98**
Pavia Centre	w/o RS	ClearCLIP	87.02	53.28	81.33
		SegEarth-OV	85.64	49.51	79.00
		Ours	**88.33**	**58.48**	**83.43**
	RS	ClearCLIP	88.30	55.94	83.24
		SegEarth-OV	88.24	55.79	82.91
		Ours	**90.62**	**69.63**	**86.75**
AeroRIT	w/o RS	ClearCLIP	69.47	64.82	54.28
		SegEarth-OV	70.34	59.02	54.48
		Ours	**80.11**	**78.81**	**71.10**
	RS	ClearCLIP	74.52	62.48	60.65
		SegEarth-OV	73.66	58.93	59.07
		Ours	**84.11**	**81.92**	**76.51**

TABLE VIII

ABLATION RESULTS OF DIFFERENT RESOLUTION SCALE COMBINATIONS ON THE AERORIT DATASET

Scale Factor	**OA (%)**	**AA (%)**	**Kappa(%)**
$1\times$	80.11	78.81	71.10
$2\times$	82.61	81.20	74.40
$3\times$	80.32	79.04	70.84
$1\times, 2\times$	**84.11**	**81.92**	**76.51**
$1\times, 3\times$	82.04	79.50	73.53
$2\times, 3\times$	78.99	75.76	68.88
$1\times, 2\times, 3\times$	81.67	78.71	72.98

TABLE IX

ABLATION RESULTS ON THE TRAINING SUBSETS, WHERE R, C, AND H DENOTE THE RANDOM, CONFIDENT, AND HARD SETS, RESPECTIVELY

R	C	H	OA(%)	AA(%)	κ(%)
✓			83.79	65.68	77.61
✓	✓		86.50	67.17	81.18
✓		✓	78.39	64.90	70.79
✓	✓	✓	**90.62**	**69.63**	**86.75**

2) Scaling Strategy: We next study the influence of the RS strategy in the PLG stage. As described in the main text, the RS strategy aims to better recognize objects of different sizes by fusing prediction results from multiple image resolutions. To evaluate its effect, we compare our full method with a variant that does not use RS; that is, pseudo-labels are generated at a single resolution and directly used for subsequent training. For reference, we also report the results of ClearCLIP and SegEarth-OV with and without the RS strategy under the same setting. The quantitative results are shown in Table VII. From the table, we can observe two main trends. First, the adoption of the RS strategy leads to a clear and consistent improvement for all three methods (ClearCLIP, SegEarth-OV, and our approach) across most metrics. This indicates that multiscale fusion is a general and effective technique for enhancing CLIP-based dense prediction in remote sensing. Second, among all variants, our full method with RS achieves the best classification performance on all evaluated datasets, which further supports the effectiveness of combining RS with spectral NLL. These results are closely related to the fixed receptive field of CLIP models. Since the spatial resolution of CLIP feature maps is limited, a single input scale cannot handle both small and large objects equally well. Higher input resolutions are more suitable for recognizing small objects and thin structures (e.g., cars, narrow roads, small buildings), because these objects occupy more pixels and are less likely to vanish during feature extraction. On the other hand, lower resolutions are better at capturing global context and large homogeneous regions (e.g., water bodies, farmland, large roofs), since they force the model to focus on coarse structures and reduce the influence of local noise. By averaging prediction results from different resolutions, the RS strategy allows us to benefit from both types of information, leading to more reliable and stable pseudo-labels. These improved pseudo-labels then serve as a stronger supervision signal in the later NLL stage.

Furthermore, to investigate the optimal configuration for the scaling factor, we conduct an ablation study on the AeroRIT dataset, as shown in Table VIII. Combining the $1\times$ and $2\times$ scales yields the optimal classification performance. This specific combination is highly effective because it seamlessly balances the recognition of both large and small objects discussed above. As evidenced by the empirical results, relying solely on the $1\times$ scale struggles to identify small objects and fine-grained structures due to limited spatial resolution. Conversely, using only the $2\times$ scale improves the OA by recovering these fine-grained details, but it still falls short of the optimal performance because the enlarged patches lose the broader global context necessary for recognizing large continuous structures. Fusing these two scales successfully captures these complementary spatial details. In addition, further increasing the scale (e.g., incorporating a $3\times$ scale) degrades the overall performance, as excessive upsampling restricts the model's receptive field to overly localized patches, fragmenting larger semantic structures and introducing interpolation noise.

3) Label Refinement Strategy: In this section, we analyze the effect of the proposed label refinement strategy, which relies on random, confident, and hard sets, as well as soft labels obtained from GMMs. We conduct ablation experiments on Pavia Center dataset and report the results in Table IX. In these experiments, we gradually add different subsets into the training process to see how each component contributes to the final performance.

When we train the spectral classifier using only the random set with hard labels (i.e., the pseudo-labels given by SegEarth-OV), the model already achieves a reasonable level of performance, because the random set distills the coarse semantic prior from the PLG stage. However, the presence of label noise still limits the classification accuracy, especially for classes with strong confusion or limited training samples. When we include the confident set with soft labels in the training, the performance improves noticeably. The confident set contains pixels whose predictions are both consistent and reliable according to the BvSB criterion and the GMM

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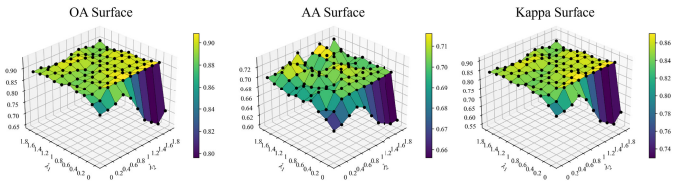


Fig. 7. Sensitivity analysis of the weighting factors λ_1 and λ_2 on Pavia Center dataset. The 3-D surfaces illustrate the variations in OA, AA, and kappa coefficient.

separation. The use of soft labels for these samples provides more flexible supervision than one-hot labels, allowing the model to better adapt to subtle class boundaries while still focusing on high-confidence information.

The inclusion of the hard set brings a further, although sometimes smaller, gain. Samples in the hard set are more likely to be mislabeled, but they often lie near decision boundaries or in mixed regions, and thus carry important information about class overlap and transition zones. By assigning soft labels to these samples and using a suitable weight in the loss function, we allow the model to explore these difficult regions without overfitting to incorrect labels. This helps the model to refine its decision boundaries and improve the classification of ambiguous pixels. It is important to note, however, that introducing the hard set alone (e.g., an R + H setting) can degrade the OA. Because these samples are intentionally selected from low-confidence regions, they contain a substantially higher noise ratio than those in the confident set. When the hard set is introduced without the confident set, the reliable anchors needed to estimate the class-wise GMM distributions are missing. The model is forced to learn from a subset with a much higher noise level, which weakens the supervision signal and may introduce confirmation bias. Therefore, the role of the confident set is crucial: when both sets are used together, the confident set stabilizes the class distributions with trustworthy soft labels, while the hard set contributes the aforementioned boundary and difficult samples, leading to the best overall result. The overall results in Table IX show that the addition of training samples with soft labels from both the confident and hard sets significantly improves performance compared to using only the random set.

Furthermore, to systematically evaluate the impact of the weighting factors λ_1 and λ_2 in the joint loss function [see (10)], we conducted a comprehensive 2-D grid-search sensitivity analysis on Pavia Center dataset. As illustrated in Fig. 7 for OA, AA, and Kappa coefficient, the model's performance exhibits a broad, stable plateau when both λ_1 and λ_2 fall within the range of approximately 0.2 to 1.6. This wide plateau demonstrates that our framework is highly robust to hyperparameter variations and does not rely on exhaustive, dataset-specific tuning. Conversely, the sharp decline in classification performance when λ_1 and λ_2 approach 0.0 mathematically validates our core methodological design, proving that the active incorporation of both the confident and hard sets is essential for effectively correcting the initial pseudo-label noise.

To further isolate the contribution of the GMM-based label refinement module, we additionally construct a *without-GMM* variant. After the warmup phase, we remove the GMM-based label refinement supervision by returning empty clean and rectified refinement masks, so that the model is trained only with the original CLIP pseudo-labels through the random-set loss branch while keeping the rest of the training pipeline unchanged. We conducted the *without-GMM* ablation experiment on the AeroRIT dataset, and the detailed results are reported in Table X. Although the without-GMM variant performs reasonably well and is even slightly better on a few individual classes, the full model achieves notably higher OA, AA, and κ. This confirms that the GMM-based refinement provides useful additional supervision beyond the original CLIP pseudo-labels.

TABLE X

CONTROLLED ABLATION OF THE GMM-BASED LABEL REFINEMENT ON THE AERORIT DATASET. THE WITHOUT-GMM VARIANT KEEPS THE REMAINING TRAINING PIPELINE UNCHANGED AND USES ONLY THE ORIGINAL SEGEARTH-OV PSEUDO-LABELS THROUGH THE RANDOM-SET LOSS AFTER WARMUP

Method	Buildings	Water	Cars	Vegetation	Soil	OA(%)	AA(%)	κ(%)
w/o GMM	85.92	76.50	**80.47**	**97.78**	51.24	80.22	78.38	70.54
Ours	**89.86**	**82.37**	77.31	95.98	**64.08**	**84.11**	**81.92**	**76.51**

TABLE XI

EFFECT OF CHANNEL-ORDER PERMUTATIONS ON PSEUDO-LABEL QUALITY AND FINAL CLASSIFICATION PERFORMANCE ON AERORIT. THE SAME THREE WAVELENGTHS ARE USED IN ALL SETTINGS, WHILE ONLY THEIR ASSIGNMENT TO THE R/G/B CHANNELS IS CHANGED. THE CANONICAL RGB ORDER ACHIEVES THE BEST RESULTS

Channel order (R, G, B)	Pseudo OA	Pseudo AA	Pseudo κ	OA	AA	κ
RGB $(655, 553, 451)$	**73.66**	**58.93**	**59.07**	**84.11**	**81.92**	**76.51**
RBG $(655, 451, 553)$	43.59	42.08	21.14	58.85	62.39	44.67
GRB $(553, 655, 451)$	41.51	43.89	14.70	34.91	51.20	11.71
GBR $(553, 451, 655)$	37.27	32.61	12.11	41.29	51.24	21.95
BRG $(451, 655, 553)$	39.08	41.82	16.91	42.49	51.45	27.92
BGR $(451, 553, 655)$	65.53	44.85	46.57	77.29	71.10	66.76

4) Effect of Spectral Wavelength Alignment for RGB Synthesis: This section aims to empirically validate the necessity of physically aligning specific hyperspectral wavelengths with standard RGB input channels to effectively leverage the semantic priors of a pretrained VLM. In the PLG stage, we synthesize a three-channel image from the original HSI using target bands at 655, 553, and 451 nm. These wavelengths were selected because they precisely align with the physical center wavelengths of standard optical imaging, purposely minimizing the domain gap between our HSI data and the natural RGB images the VLM was originally trained on. To evaluate the impact of this physical alignment, we conducted an ablation study on the AeroRIT dataset by fixing the interpolated bands at 655, 553, and 451 nm and systematically permuting their assignments across the VLM's three input channels. The results, presented in Table XI, demonstrate that the canonical RGB order (655, 553, and 451 nm) achieves the highest performance across all metrics. In contrast, altering the channel assignment causes severe degradation in both PLG and final classification. For instance, the GRB permutation drastically drops the final OA to 34.91%. These results confirm that matching these specific physical wavelengths is important

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TABLE XII

EFFECT OF DIFFERENT SPATIAL INTERPOLATION METHODS USED IN THE RS STRATEGY ON FINAL CLASSIFICATION PERFORMANCE ON THE CHIKUSEI DATASET

Interpolation Method	OA (%)	AA (%)	κ (%)
Nearest	86.87	**85.81**	83.00
Bilinear	80.43	82.60	74.86
Bicubic	**88.33**	85.11	**84.98**
Spline (5th-order)	86.28	84.50	82.37

TABLE XIII

STATISTICAL SIGNIFICANCE ANALYSIS OF OUR METHOD AGAINST ALL COMPETING METHODS ON THE CHIKUSEI DATASET USING PAIRED T-TESTS BASED ON CLASS-WISE ACCURACIES. A RESULT IS CONSIDERED STATISTICALLY SIGNIFICANT WHEN $p < 0.05$

Compared method	p-value	Significant
CLIP [20]	9.63×10^{-4}	Yes
MaskCLIP [38]	2.67×10^{-2}	Yes
SCLIP [39]	2.60×10^{-2}	Yes
GEM [40]	2.30×10^{-2}	Yes
ClearCLIP [41]	3.12×10^{-2}	Yes
SegEarth-OV [25]	5.33×10^{-2}	No
CTF-SSCL [44]	5.63×10^{-4}	Yes
DSNet [45]	4.37×10^{-3}	Yes
HyperTaFOR [46]	1.90×10^{-5}	Yes
MCTGCL [47]	2.02×10^{-3}	Yes
Semi-HCN [48]	8.79×10^{-3}	Yes

for generating high-quality pseudo labels and achieving the best final performance.

5) Impact of Spatial Interpolation Methods: This section investigates the impact of different spatial interpolation methods used in the RS strategy during the PLG stage. Specifically, we evaluate different image resampling operators for the upsampling and downsampling operations in (9), including nearest-neighbor, bilinear, bicubic, and fifth-order spline interpolation, on the Chikusei dataset. As detailed in Table XII, while the choice of spatial interpolation method influences the final classification performance, the overall fluctuations remain relatively constrained. Among the evaluated methods, bicubic interpolation achieves the best performance. We attribute this to its favorable balance between preserving local spatial structures and suppressing interpolation artifacts. Compared with nearest-neighbor interpolation, bicubic interpolation produces smoother and more coherent prediction maps with fewer blocky artifacts. Compared with bilinear interpolation, it better preserves object boundaries and fine structures, which are important for recognizing small objects and thin regions in remote sensing scenes. The fifth-order spline interpolation also yields competitive results, but remains slightly inferior to bicubic under our setting. Overall, these results indicate that the proposed framework is reasonably robust to the specific spatial interpolation operator used in the RS stage, while bicubic interpolation provides the most effective choice for our multiscale fusion strategy.

6) Statistical Significance Analysis: To further evaluate the reliability of the performance gains, we conduct paired t-tests on the Chikusei dataset based on the class-wise accuracies, and the results are reported in Table XIII. It can be observed that our method achieves statistically significant improvements over most competing methods with $p < 0.05$. For SegEarth-OV, the obtained p-value is 5.33×10^{-2}, which is slightly above the conventional significance threshold. We would like to note that the paired t-test here is conducted on only a small number of class-wise samples, which may limit its statistical power. Nevertheless, our method still shows clear advantages on the standard evaluation metrics. Specifically, compared with SegEarth-OV, our method improves OA from 75.54% to 88.33%, AA from 59.96% to 85.11%, and κ from 67.49% to 84.98% on Chikusei. These consistent gains on the main hyperspectral classification metrics further demonstrate the effectiveness of the proposed method.

TABLE XIV

COMPARISON BETWEEN THE PROPOSED ZERO-SHOT METHOD AND SUPERVISED MAMBAHSI ON THE CHIKUSEI DATASET

Method	Setting	OA(%)	AA(%)	κ(%)
MambaHSI	Supervised (1 label)	60.29	60.76	49.03
MambaHSI	Supervised (3 labels)	72.93	73.68	65.22
MambaHSI	Supervised (5 labels)	78.81	81.04	72.85
MambaHSI	Supervised (10 labels)	**88.81**	**88.94**	**85.56**
Ours	Zero-shot	88.33	85.11	84.98

TABLE XV

COMPARISON OF COMPUTATIONAL EFFICIENCY (MODEL COMPLEXITY, TRAINING TIME, AND INFERENCE TIME) ON PAVIA CENTER DATASET

Method	**Parameters (M)**	**FLOPs (G)**	**Training Time (s)**	**Inference Time (s)**
CLIP	149.62	1986.17	-	0.40
MaskCLIP	149.62	1982.93	-	0.41
SCLIP	149.62	1989.44	-	0.40
GEM	149.62	2184.29	-	0.68
ClearCLIP	149.62	1885.70	-	0.39
SegEarth-OV	149.91	5119.17	-	1.23
CTF-SSCL [44]	0.36	0.01	34.33	2.73
DSNet [45]	12.58	0.01	65.87	27.09
HyperTaFOR [46]	0.09	0.03	59.72	6.58
MCTGCL [47]	0.24	0.06	35.33	7.65
Semi-HCN [48]	0.07	69.46	45.51	3.56
Ours	**149.95**	**26782.26**	**409.16**	**0.25**

7) Comparison With Supervised MambaHSI: To further clarify the zero-shot capability of the proposed method, we additionally compare it with MambaHSI under supervised settings on the Chikusei dataset, and the results are reported in Table XIV. Different from MambaHSI, which relies on labeled training samples, our method performs classification in a zero-shot manner without using any task-specific annotations. As shown in Table XIV, the proposed method achieves OA/AA/κ of 88.33%/85.11%/84.98% in the zero-shot setting. It substantially outperforms supervised MambaHSI when only one, three, or five labeled samples are available. Specifically, compared with MambaHSI under the five-label supervised setting, our method improves OA, AA, and κ by 9.52, 4.07, and 12.13 percentage points, respectively. Even when compared with the ten-label supervised setting, our method remains highly competitive, with absolute differences of only 0.48, 3.83, and 0.58 percentage points in OA, AA, and κ, respectively. These results demonstrate that the proposed method can achieve competitive classification performance without annotated training data, highlighting its strong zero-shot capability and favorable label efficiency.

8) Computational Efficiency: To provide a comprehensive evaluation of our framework, we compare the computational efficiency of SPECIAL against both the CLIP-based zero-shot baselines and several recent HSI classification methods,

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including CTF-SSCL [44], DSNet [45], HyperTaFOR [46], MCTGCL [47], and Semi-HCN [48]. The comparison specifically evaluates model complexity (Parameters and FLOPs), training time, and inference time on Pavia Center dataset. The quantitative results are summarized in Table XV. As shown in the table, the total parameter count for our SPECIAL framework is approximately 149.95 M. It is worth noting that the vast majority of these parameters (149.91 M) and computational overhead belong to the CLIP-based PLG stage. In contrast, the MambaHSI spectral classifier utilized in the NLL stage is lightweight, containing only about 33.7-K parameters. Consequently, the substantial FLOPs burden primarily impacts the training process; during the testing phase, the framework relies exclusively on the lightweight MambaHSI classifier without invoking the heavy CLIP model, ensuring rapid inference. In terms of execution, the total training time for our method is approximately 409.16 s. More importantly, a single full-image inference takes merely 0.25 s. Compared to the recent DL baselines—which often require significantly longer inference times (e.g., 27.09 s for DSNet)—our framework remains highly efficient during the operational phase, demonstrating its strong potential for practical, real-world deployment.

V. Conclusion

In this article, we propose a novel zero-shot HSI classification framework, SPECIAL, which leverages the capabilities of CLIP for pixel-level classification without the need for manual annotation. The framework consists of two stages, including CLIP-based PLG and NLL. In the PLG stage, the framework interpolates HSI data to obtain RGB bands and uses CLIP for initial classification, generating noisy pseudo-labels and confidence scores. A scaling strategy is proposed to improve the quality of the pseudo-labels. In the NLL stage, spectral information and a label refinement strategy are incorporated to alleviate the issue of label noise and further enhance accuracy. Experiments on three datasets demonstrate that SPECIAL achieves competitive and often superior performance over existing approaches in zero-shot HSI classification.

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